

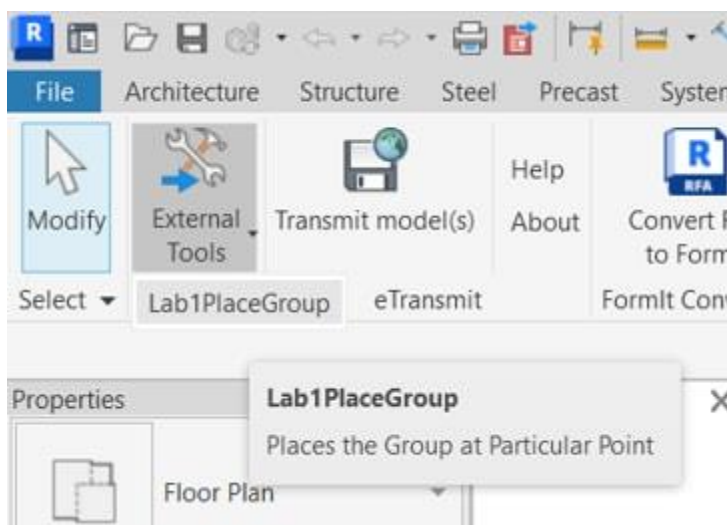
[My First Revit Plug-in Overview](#) >> [Lesson 1](#) >> [Lesson 2](#) >> [Lesson 3](#)>> [Lesson 4](#) >> [Lesson 5](#) >> [Lesson 6](#) >> [Lesson 7](#) >> [Lesson 8](#)

My First Plug-in Training

[This page has been updated to work with Revit 2025 and Revit 2026. If you are interested in previous versions, (i.e., Revit 2023&2024), you can download the pdf version of the training materials from [here](#)]

Lesson 1: The Basic Plug-in

In this lesson you will create your very first basic Autodesk Revit plug-in for copying groups selected by the user to a specified location.



[View Now](#)

Provide Feedback: Please provide feedback about this Revit Training or this lesson via email: myfirstplugin@autodesk.com

Lesson Downloads

 [lesson 1_revit_projects.zip](#)

 [Sample Revit Project File: Hotel.rvt](#)

Steps to Create Your First Plug-in

1. **Launch the Visual Studio development environment:**
Open Visual Studio using the Windows **Start** menu, selecting **All Programs**, and then **Microsoft Visual Studio**.

Supported .NET version for Revit version is as follows:

- Revit 2026/2025 - .NET 8.0 (Long Term Support)

Note: previous versions of Revit 2023 and 2024 uses .NET 4.8. If you are interested in the training materials using previous versions of Revit, you can download the pdf version of the materials from [here](#)],

You can find the developer guide for the versions here:

Revit

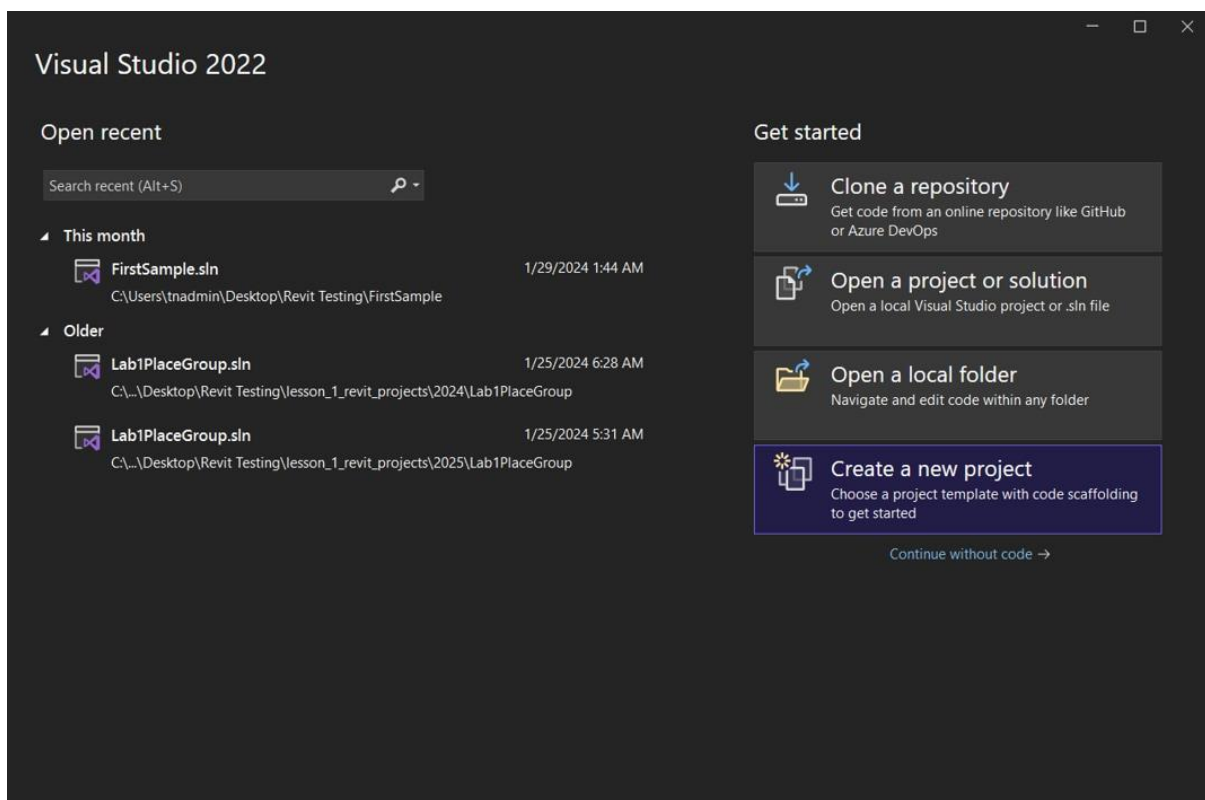
2025: https://help.autodesk.com/view/RVT/2025/ENU/?guid=Revit_API_Revit_API_Developers_Guide_html

Revit

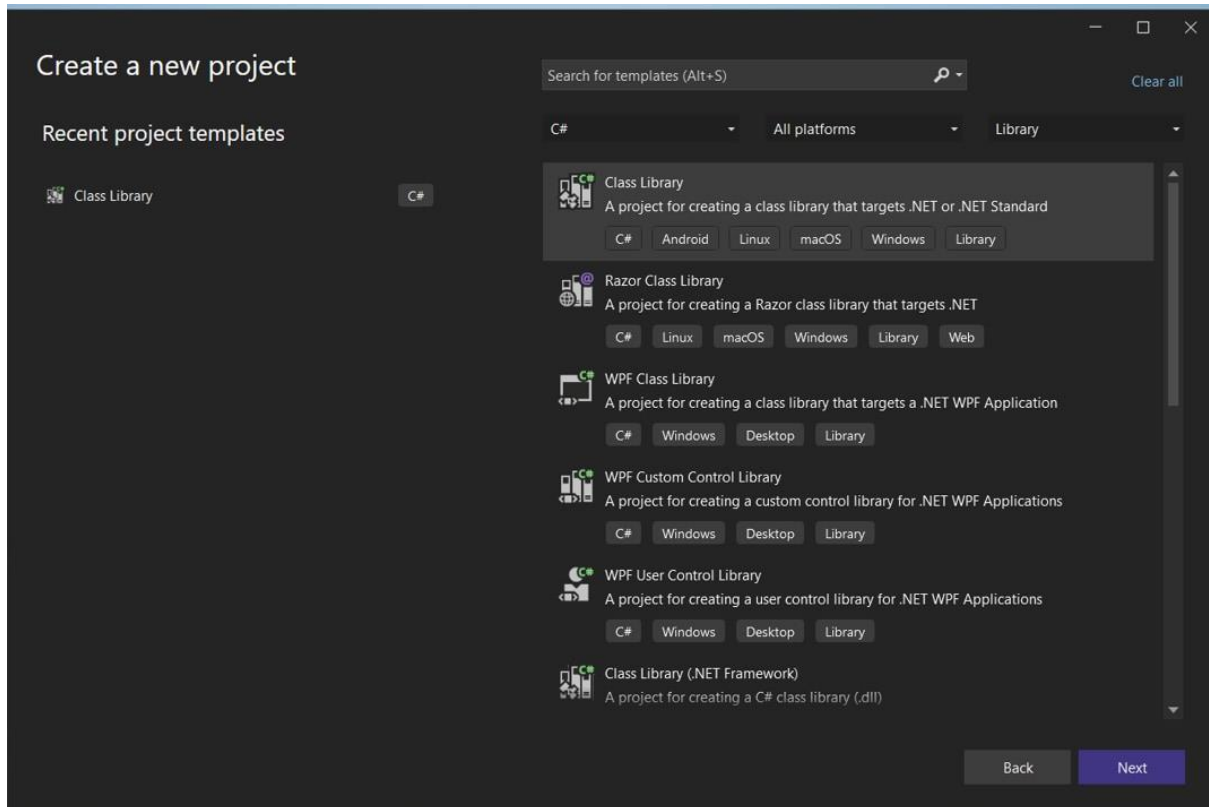
2026: https://help.autodesk.com/view/RVT/2026/ENU/?guid=Revit_API_Revit_API_Developers_Guide_html

2. Create a class library project:

1. Open Visual Studio.
2. Navigate to the "Get started" section. Click on "Create a new project".

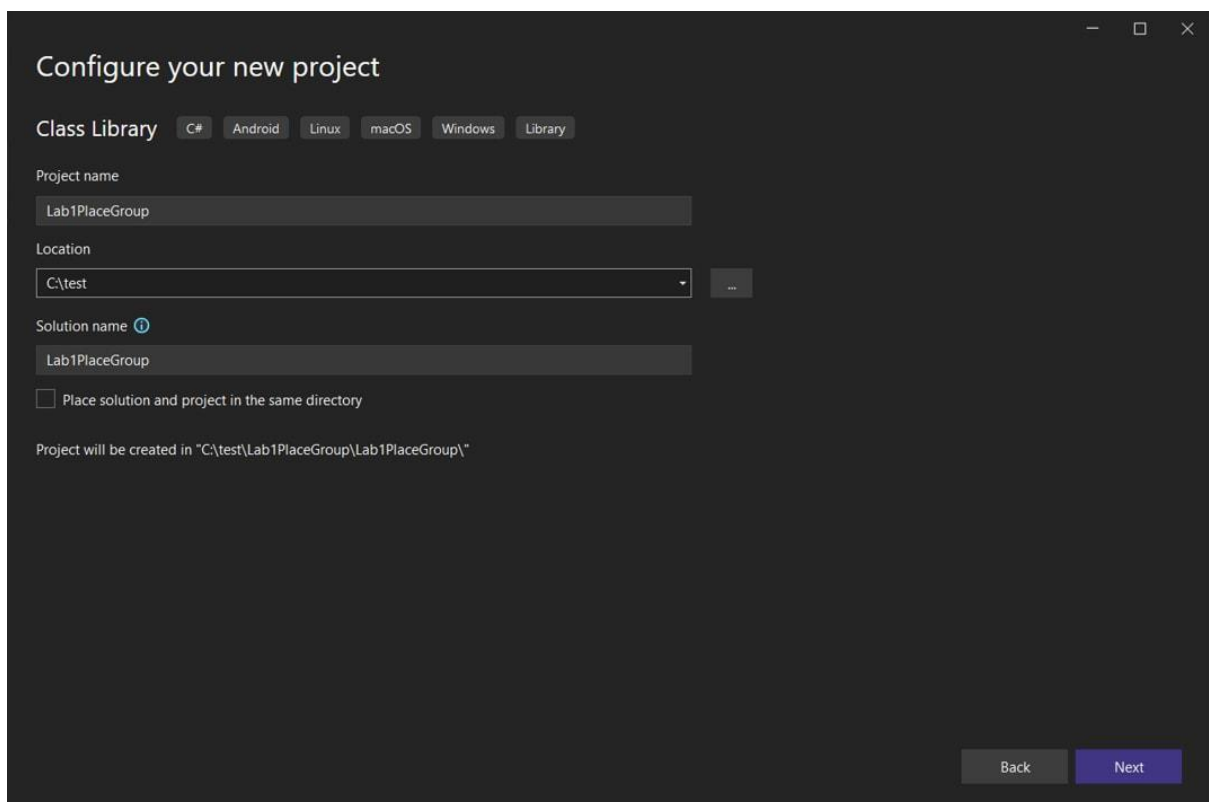


- c. Select "C#" from languages.
- d. Choose "All platforms".
- e. Select "Library" as the project type.

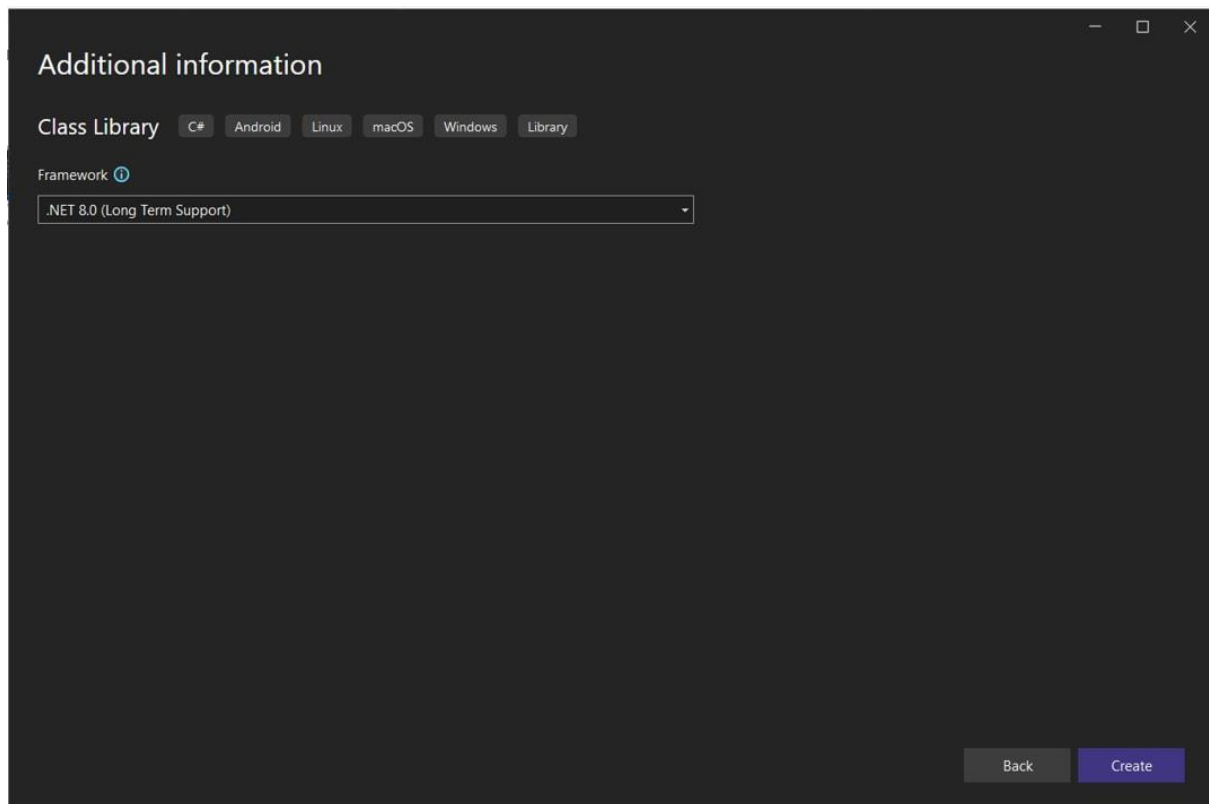


f. Now, choose "Class Library – A project for creating a class library that targets .NET or .NET Standard" from the available options. Click on the "Next" button.

g. Enter "Lab1PlaceGroup" as the Project Name.



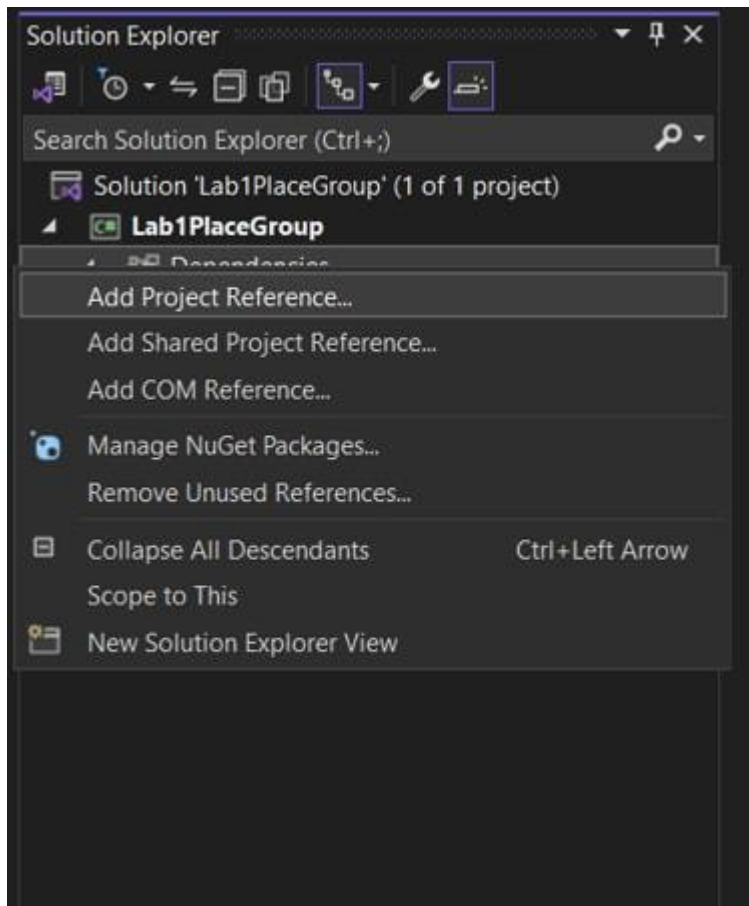
- h. Type "C:\test" in the Location box.
- i. Click on the "Next" button.
- j. In the "Framework" section, choose ".NET 8.0 (Long Term Support)". Finally, click on the "Create" button to create the project.



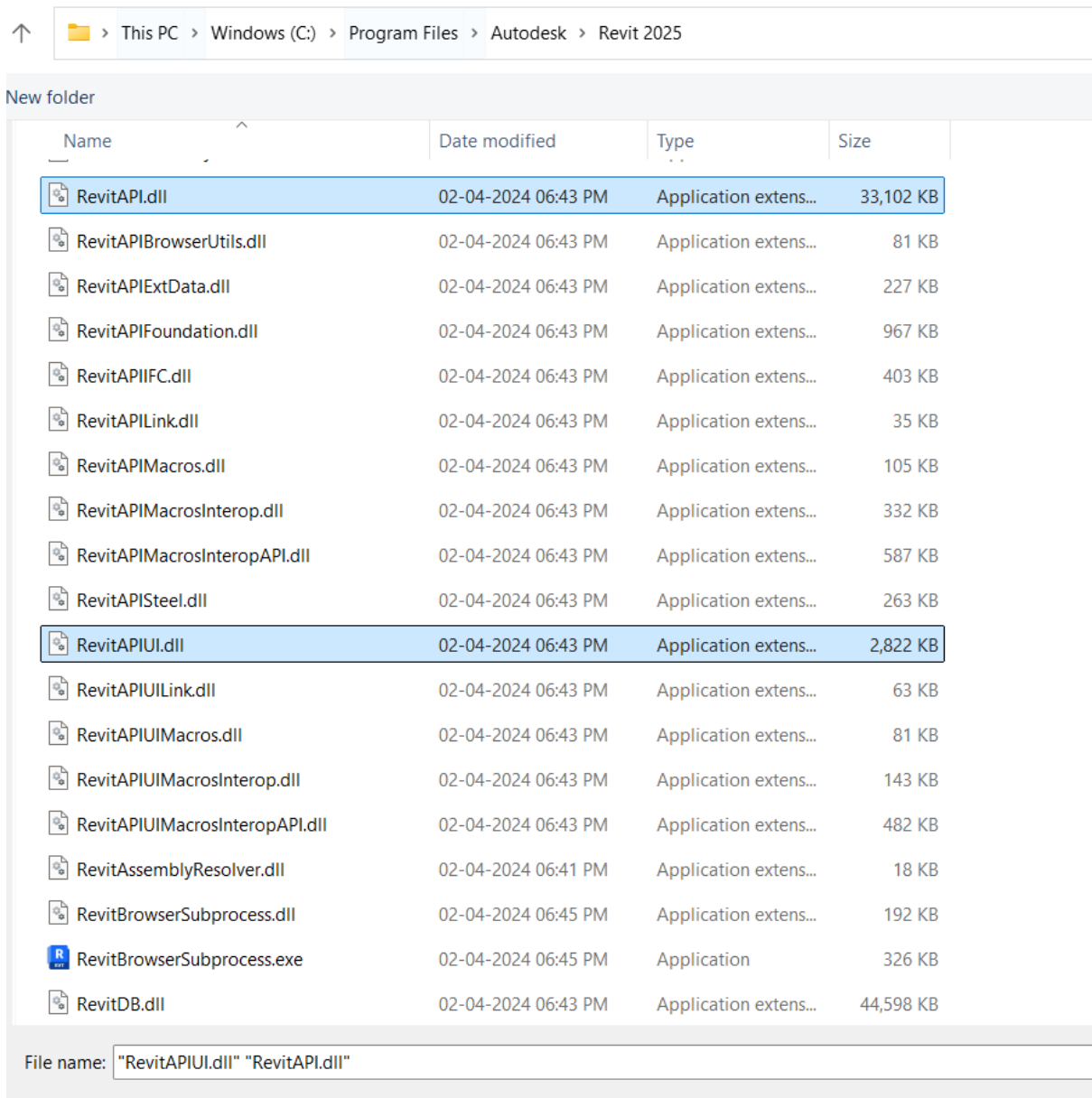
3. **Add references:**

In the "View" menu, you'll see "Solution Explorer". Click on it. This action should open the Solution Explorer window.

Within the Solution Explorer, right-click "Dependencies" and select "Add Project Reference...".



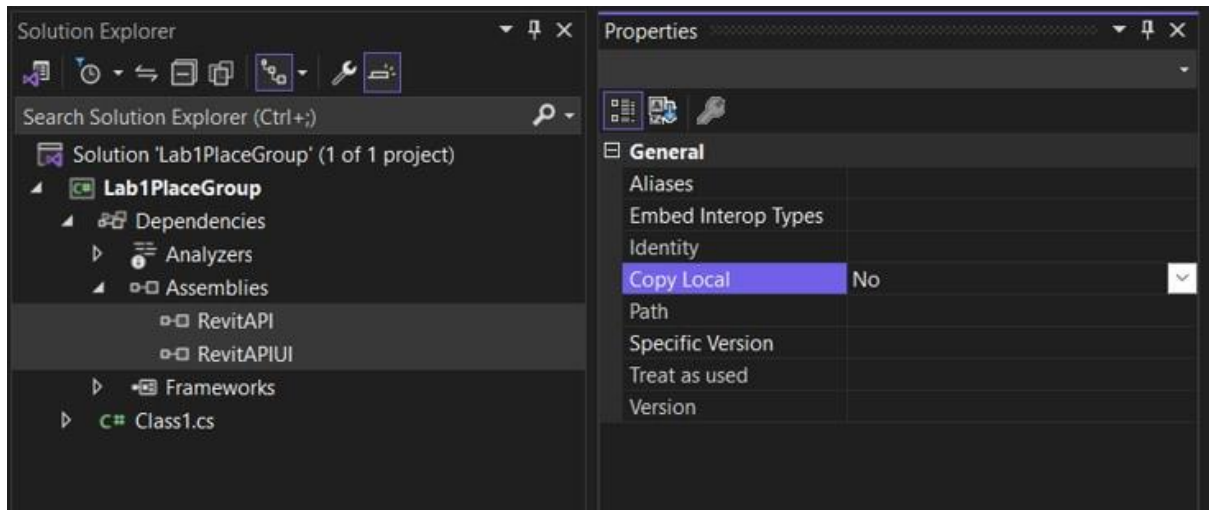
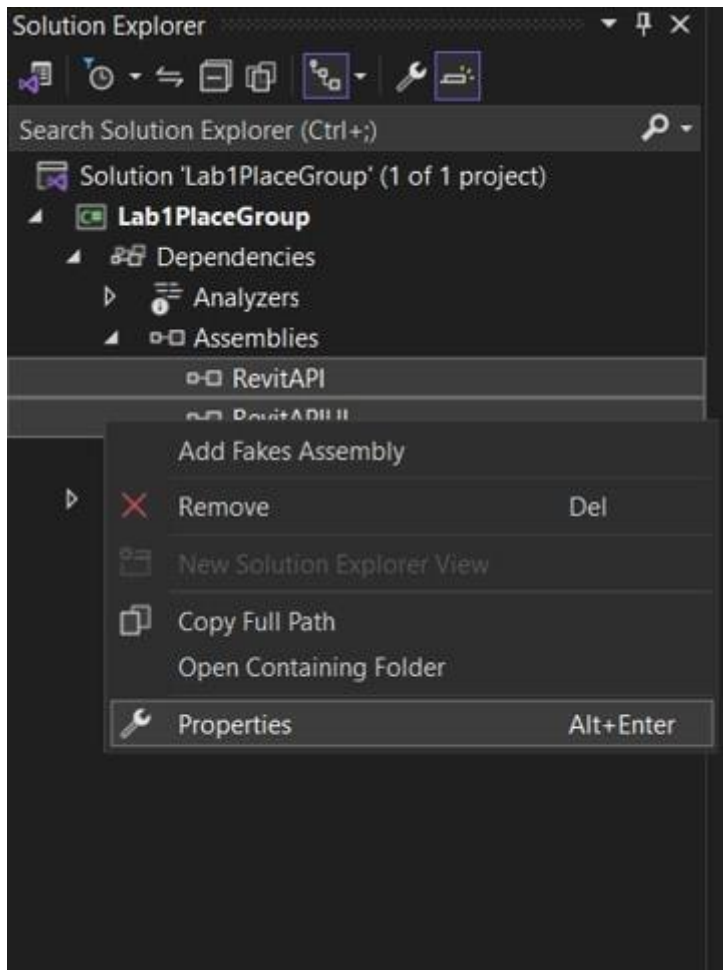
4. Click the Browse tab and in the Add Reference dialog and browse to the Revit product installation sub-folder. (The sub-folder path depends on where you have installed Revit 20xx. The default path is C:\Program Files\Autodesk\Revit 20xx*).



You will add two reference files from this folder. Select **RevitAPI.dll**, hold the Ctrl key and select **RevitAPIUI.dll**, and then click OK. Now the two interface DLL files are referenced in your project. All the Revit APIs are exposed by these interface files and your project can use all of those available APIs from them.

5. Set the referenced files' Copy Local property value

In the **Solution Explorer** window you saw in step 4, right click **RevitAPI** under **Assemblies** node and choose properties. In the **Properties** window, click **Copy Local** property, and then click the drop-down list, select **No**. Repeat the same steps to change **RevitAPIUI**'s **Copy Local** property value to **No**.



6. Add the code:

Double click **Class1.cs** in the **Solution Explorer** window to show the code-editing window. Delete everything in this window and then type the following C# code. To get the full experience of developing with Visual Studio – including the use of features such as IntelliSense – we recommend you type the code from this guide rather than copying and pasting it. That said, if

constrained for time you can also copy and paste into the Visual Studio code window: although this reduces the experience you gain from working with the code directly.

```
using Autodesk.Revit.DB;
```

```
using Autodesk.Revit.Attributes;
```

```
using Autodesk.Revit.UI;
```

```
using Autodesk.Revit.UI.Selection;
```

```
namespace Lab1PlaceGroup
```

```
{
```

```
    [Transaction(TransactionMode.Manual)]
```

```
    public class Class1 : IExternalCommand
```

```
    {
```

```
        public Result Execute(ExternalCommandData commandData, ref string message,  
        ElementSet elements)
```

```
        {
```

```
            //Get application and document objects
```

```
            UIApplication uiapp = commandData.Application;
```

```
            Document doc = uiapp.ActiveUIDocument.Document;
```

```
            //Define a reference Object to accept the pick result
```

```
            Reference pickedref;
```

```
            //Pick a group
```

```
            Selection sel = uiapp.ActiveUIDocument.Selection;
```

```
            pickedref = sel.PickObject(ObjectType.Element, "Please select a group");
```

```
            Element elem = doc.GetElement(pickedref);
```

```
            Group? group = elem as Group;
```

```
            //Pick point
```

```
            XYZ point = sel.PickPoint("Please pick a point to place group");
```



```
//Place the group  
  
Transaction trans = new Transaction(doc);  
  
trans.Start("Lab");  
  
doc.Create.PlaceGroup(point, group?.GroupType);  
  
trans.Commit();  
  
return Result.Succeeded;  
}  
}  
}
```

Don't worry about the details of the code for now, you'll come back to this shortly in the next couple of lessons.

7. **Save the file:**

On the **File** menu, click **Save All**.

8. **Build the project:**

The code you have written is in human readable form. To make the code readable by a computer, you will need to translate it or “build” it.

Inside Visual Studio, in the Build menu, click Build Solution to compile and build your plug-in. Build Success message shows in status bar of the Visual Studio window if the code is successfully built.

```

4  [using Autodesk.Revit.UI.Selection;
5
6  namespace Lab1PlaceGroup
7  {
8      [Transaction(TransactionMode.Manual)]
9      public class Class1 : IExternalCommand
10     {
11         public Result Execute(ExternalCommandData commandData, ref string message, ElementSet elements)
12         {
13             //Get application and document objects
14             UIApplication uiapp = commandData.Application;
15             Document doc = uiapp.ActiveUIDocument.Document;
16
17             //Define a reference Object to accept the pick result
18             Reference pickedref;
19
20             //Pick a group
21             Selection sel = uiapp.ActiveUIDocument.Selection;
22             pickedref = sel.PickObject(ObjectType.Element, "Please select a group");
23
24             Element elem = doc.GetElement(pickedref);
25             Group? group = elem as Group;
26
27             //Pick point
28             XYZ point = sel.PickPoint("Please pick a point to place group");
29
30             //Place the group
31             Transaction trans = new Transaction(doc);
32             trans.Start("Lab");
33             doc.Create.PlaceGroup(point, group?.GroupType);
34             trans.Commit();
35
36             return Result.Succeeded;
37         }
38     }
39 }
40

```

9. Supress Warnings

After building the solution, you will encounter numerous warnings displayed in the output window located at the bottom of the Visual Studio interface.

```

Output
Show output from: Build
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: C:\Program Files\Autodesk\Revit Preview Release\CefSharp.Wpf.dll
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: Project file item includes which caused reference "C:\Program Files\Au
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: RevitAPIUI
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: C:\Program Files\Autodesk\Revit Preview Release\Autodesk.Weave.Wpf.dll
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: Project file item includes which caused reference "C:\Program Files\Au
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: RevitAPIUI
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: C:\Program Files\Autodesk\Revit Preview Release\Autodesk.UI.Windows.WebO
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: Project file item includes which caused reference "C:\Program Files\Au
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: RevitAPIUI
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: C:\Program Files\Autodesk\Revit Preview Release\CefSharp.Core.Runtime.dl
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: Project file item includes which caused reference "C:\Program Files\Au
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: RevitAPIUI
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: C:\Program Files\Autodesk\Revit Preview Release\CefSharp.Wpf.HandHost.dl
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: Project file item includes which caused reference "C:\Program Files\Au
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: RevitAPIUI
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: There was a mismatch between the processor architecture of the project being bui
D:\Program Files\Microsoft Visual Studio\2022\Enterprise\VSBuild\Current\Bin\amd64\Microsoft.Common.CurrentVersion.targets(2382,5): warning MSB3277: There was a mismatch between the processor architecture of the project being bui
D:\test\Lab1PlaceGroup\Lab1PlaceGroup\Class1.cs(32,42,32,47): warning CS8602: Dereference of a possibly null reference.
>Lab1PlaceGroup -> C:\test\Lab1PlaceGroup\Lab1PlaceGroup\bin\Debug\net8.0\Lab1PlaceGroup.dll
>Done building project "Lab1PlaceGroup.csproj".
----- Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped -----
----- Build completed at 11:56 PM and took 00.660 seconds -----

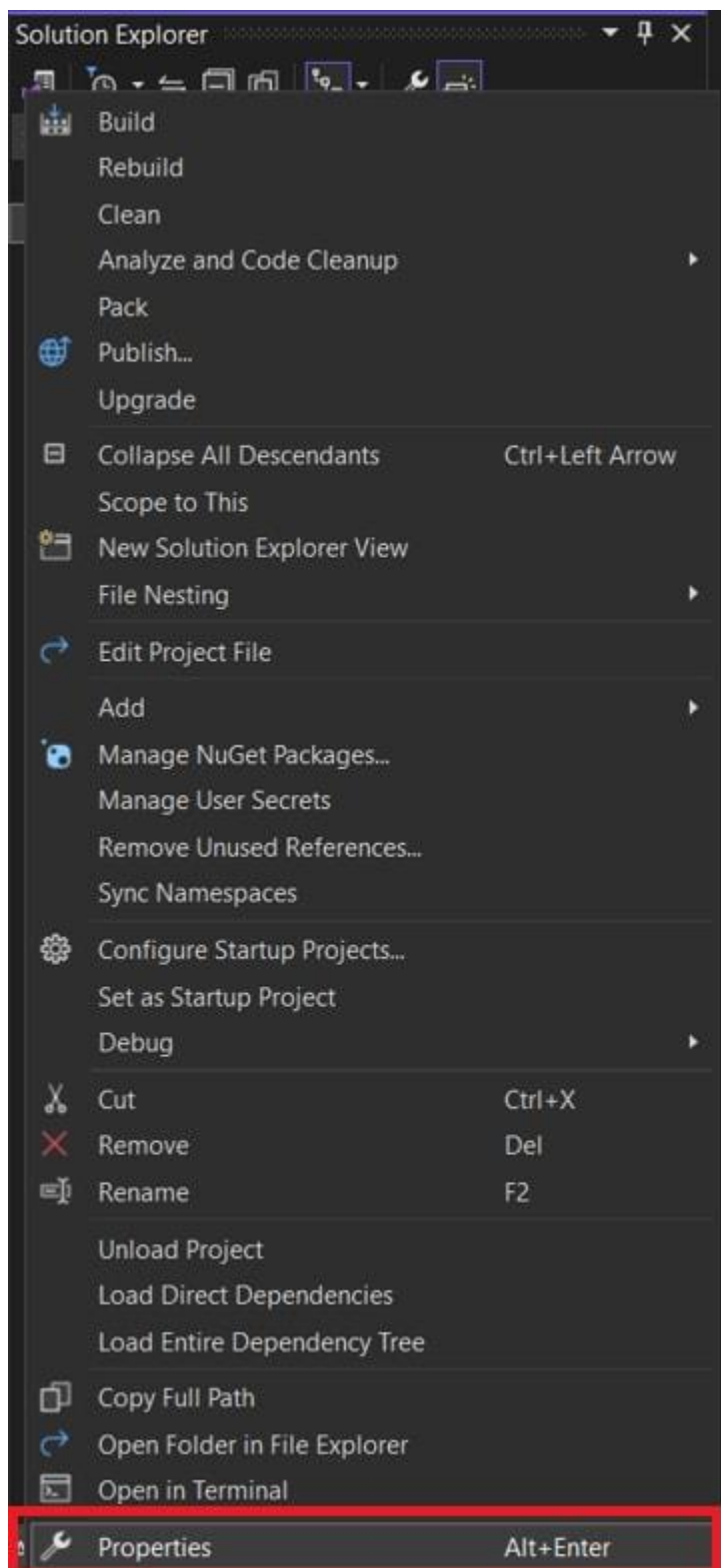
```

In this tutorial code, they are benign.

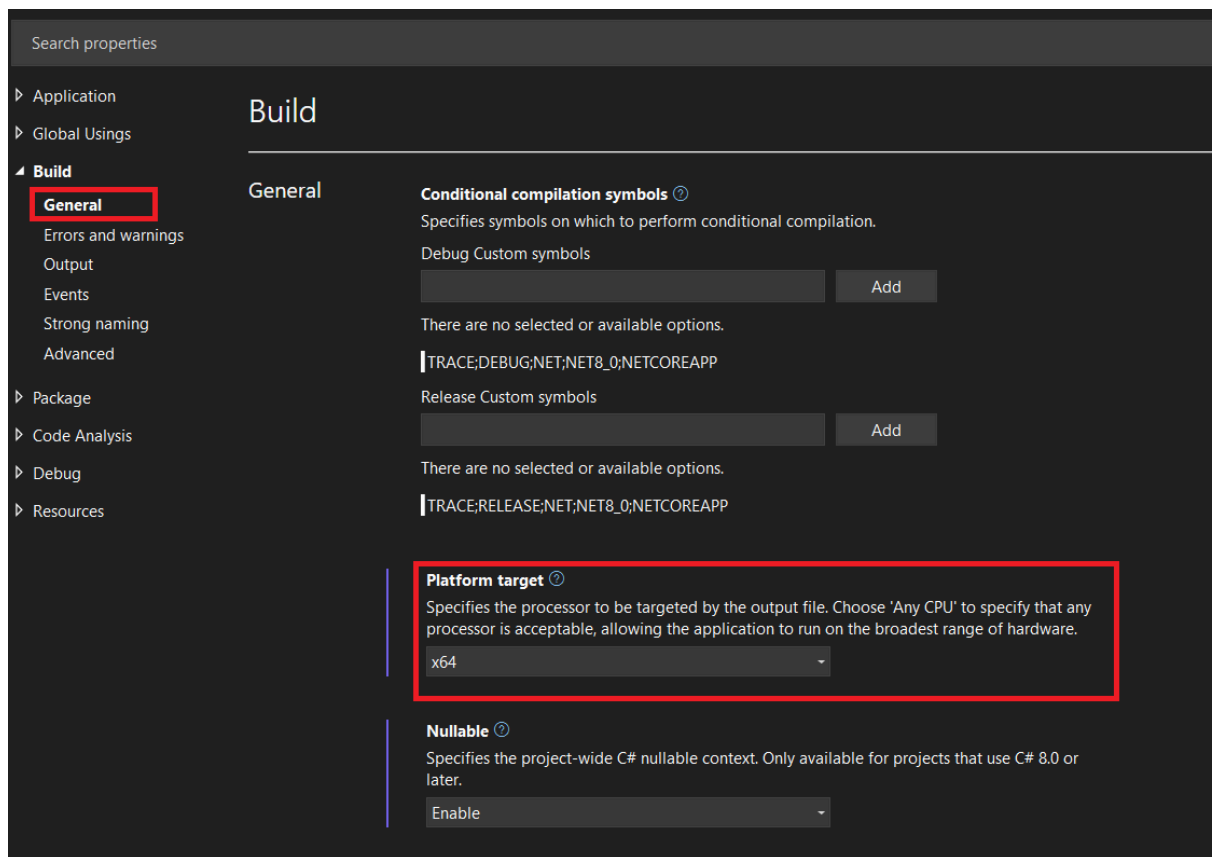
To suppress these warnings, follow these steps:

1. Navigate to the Solution Explorer.

2. Right-click on "Lab1PlaceGroup" and select "Properties."

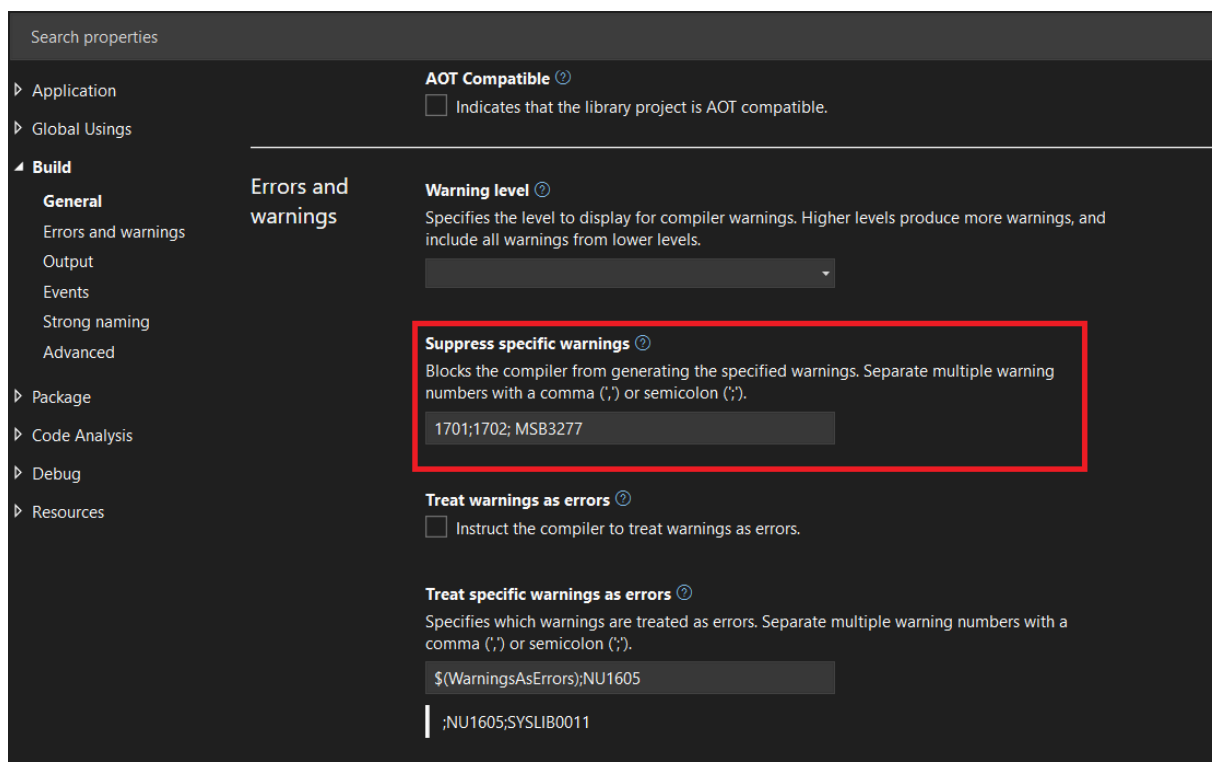


3. In the properties window, under the "Build" tab, navigate to "General."
4. Change the "Platform target" to "x64."



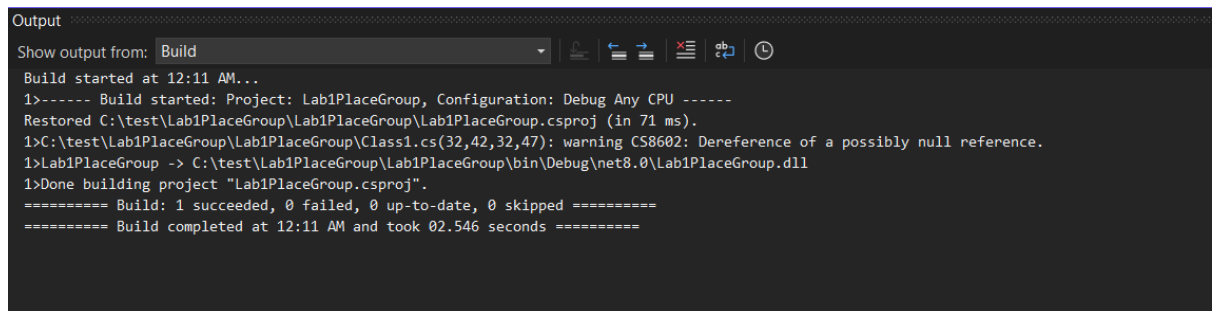
Next:

5. Still in the "Build" tab, navigate to "Errors and Warnings."
6. Add ";MSB3277" to the existing list to suppress warnings effectively.



Once these settings are configured:

7. Proceed to build the solution as instructed in step 8.



```
Output
Show output from: Build
Build started at 12:11 AM...
1>----- Build started: Project: Lab1PlaceGroup, Configuration: Debug Any CPU -----
Restored C:\test\Lab1PlaceGroup\Lab1PlaceGroup\Lab1PlaceGroup.csproj (in 71 ms).
1>C:\test\Lab1PlaceGroup\Lab1PlaceGroup\Class1.cs(32,42,32,47): warning CS8602: Dereference of a possibly null reference.
1>Lab1PlaceGroup -> C:\test\Lab1PlaceGroup\Lab1PlaceGroup\bin\Debug\net8.0\Lab1PlaceGroup.dll
1>Done building project "Lab1PlaceGroup.csproj".
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
===== Build completed at 12:11 AM and took 02.546 seconds =====
```

Upon completion of the build process, you will observe that the warnings are successfully suppressed, and the status bar of the Visual Studio window will display the message "Build succeeded."

That's it! You have just written your first plug-in for Autodesk Revit.

Before you actually work with the plug-in in Revit, you will need to do one more step, which is to write an AddIn manifest.

Writing an AddIn Manifest

An AddIn manifest is a file located in a specific location checked by Revit when the application starts. The manifest includes information used by Revit to load and run the plug-in.

1. Add the manifest code:

Start Notepad.exe from the Windows Start menu. Copy and paste the following plug-in load settings to the Notepad editor.

Note: In the add-in file, you will see a `<AddInId>GUID </AddInId>` tag. It is a GUID that represents the id of this particular application. AddInId must be unique for a given session of Revit. Autodesk recommends you generate a unique GUID for each registered application or command.

```
<?xml version="1.0" encoding="utf-8"?>
<RevitAddIns>
  <AddIn Type="Command">
    <Name>Lab1PlaceGroup</Name>
    <FullClassName>Lab1PlaceGroup.Class1</FullClassName>
    <Text>Lab1PlaceGroup</Text>
    <Description>Places the Group at Particular Point</Description>
    <VisibilityMode>AlwaysVisible</VisibilityMode>
  <Assembly>C:\test\Lab1PlaceGroup\Lab1PlaceGroup\bin\Debug\net8.0\Lab1PlaceGroup.dll</Assembly>
    <AddInId>502fe383-2648-4e98-adf8-5e6047f9dc34</AddInId>
    <VendorId>ADSK</VendorId>
    <VendorDescription>Autodesk, Inc, www.autodesk.com</VendorDescription>
  </AddIn>
</RevitAddIns>
```

Depending on the environment you are working on, you may need to change the path here to match your Lab1PlaceGroup.dll location on your computer:

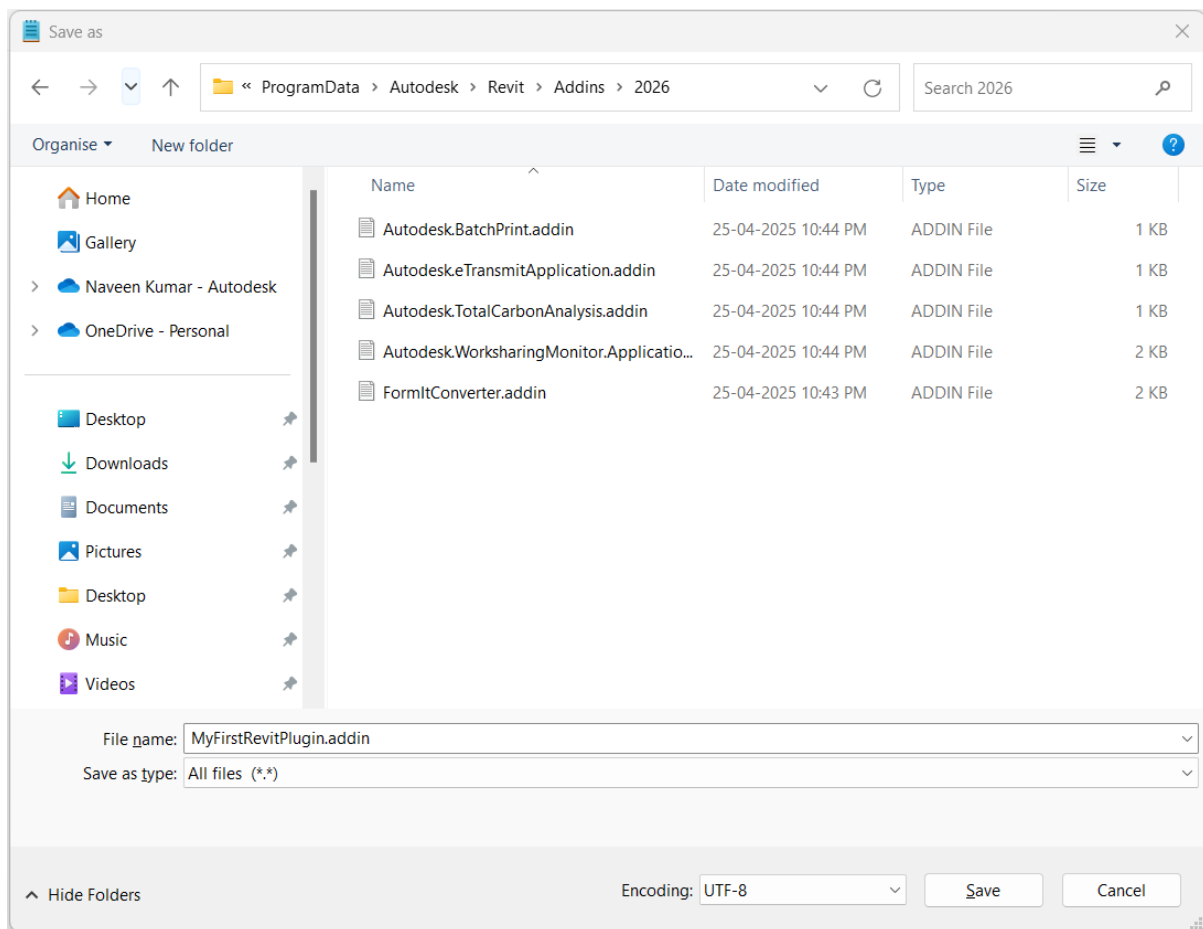
C:\test\Lab1PlaceGroup\Lab1PlaceGroup\bin\Debug\net8.0\Lab1PlaceGroup.dll

2. Save the file:

On **Notepad's File** menu, click Enter **MyFirstRevitPlugin.addin** in the **File name** box. Change **Save as type** to the **All Files** option (the file name is up to you; however, the file extension must be ".addin"). Browse to the following subfolder, and then click the **Save** button.

Under Windows 10 - C:\ProgramData\Autodesk\Revit\Addins\20xx\ (The ProgramData folder is hidden by default)

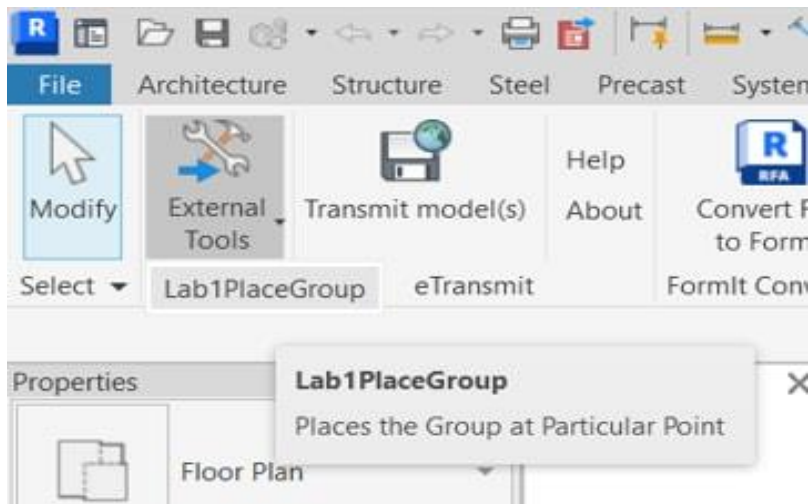
For example, here is the setting in Save As dialog in Windows 10 for Revit 2025.



3. Load your plug-in into Revit and allow the plug-in to communicate with Revit:

Start Revit. Open the sample model HOTEL.rvt.

Inside Revit on the **Add-Ins** ribbon tab, click the **External Tools** drop-down list, then click **Lab1PlaceGroup**. This will start your plug-in.



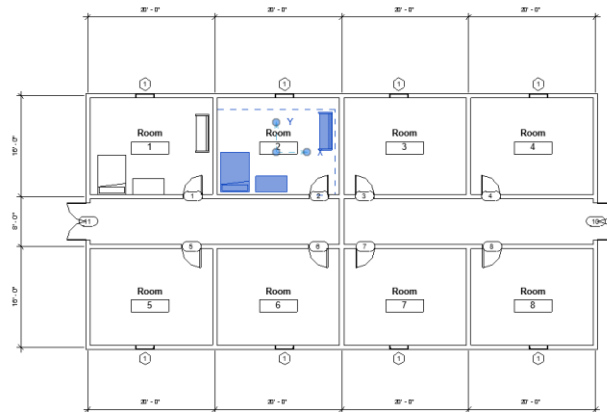
4. Work with the plug-in:

Move the cursor over **Room1** in the Revit building model. When the cursor is hovering over the furniture group, its bounding box should be highlighted as per the below picture, with a tooltip showing **Model Groups : Model Group : Group 1**. Click to select this furniture group. (Note: when highlighted the room looks very similar to the group. Please carefully select the group according to the message in the tooltip. If the room is selected, you will not see the expected result after the following step.)



NOTE: Make sure the TOOLTIP message is ModelGroups, then click to pick.

4. Pick a point in another room, for example in **Room 2**. You should see the group copied to this location. The center of the new group is the point you selected.



Congratulations! You have just written your first plug-in for Autodesk Revit. You will be reviewing the code in detail in [Lesson 3](#).

Before you move on to the next lessons, let us go back to some of the things we skipped over earlier, starting with basics concept about programming, and the benefits it can bring to your day-to-day work.

Additional Topics

Introduction to Programming

The C# code you have just executed to copy a group is only 30 lines long. Here you see a small amount of code working in a similar way to the internal Revit command, Create Similar. Software programming allows you to capture the logic of a particular functionality once and then reap the benefits over and over again, every time you want to perform this functionality.

What is Programming?

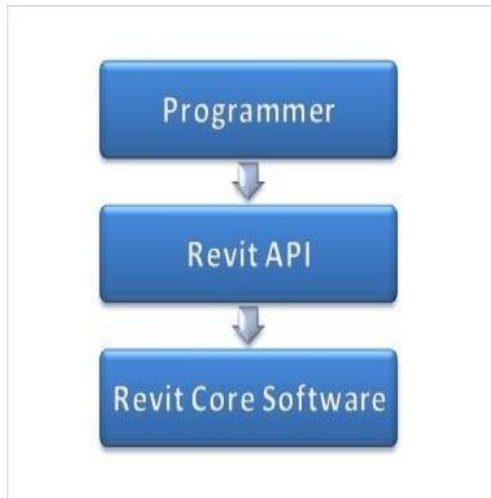
A simple answer to this question is: Computer programming is the process of creating a sequence of instructions to tell the computer to do something. You can look at your program as a sequence of instructions. During the course of the upcoming lessons, you will look at the various lines and blocks of code and look at them all in the context of being instructions for a computer.

If you were to explain what computers are to a young child, you might say: a computer is a tool which follows instructions you provide. Programming is one way of giving instructions to the computer. Internally, a computer sees these instructions encoded as a series of numbers (also called machine code). The human-readable instructions you saw at the beginning of this lesson is called source code and the computer converts these instructions to machine code which it can then read and execute. A sequence of such instructions (or code), written to perform a specific task, is called a program and a collection of such programs and related data is called a software. Autodesk Revit is one such software product.

Source code can be written in different languages, just as humans use different languages to communicate between ourselves. The language you will be using in this guide is called C# (pronounced “C-Sharp”).

What is an API?

API is the acronym for Application Programming Interface: the way a software programmer can communicate with a software product. For instance, the Revit API is the way programmers can work with Revit, and it establishes what functionality a software programmer can use within Revit. Such as the Revit API allows you to write instructions for Revit to execute one after the other.



What is a Plug-in?

A software plug-in is a type of program module (or file) that adds functionality to a software product, usually in the form of a command automating a task or some customization of the product's behavior. When you talk about a plug-in for Revit – and you will also hear the term Add-In used for this product – we mean a module containing code that makes use of the Revit API. Revit loads such plug-ins and uses them to adjust its behavior under certain conditions, such as when a particular command is executed by the user of the plug-in.

[My First Revit Plug-in Overview](#) >> [Lesson 1](#) >> [Lesson 2](#) >> [Lesson 3](#) >> [Lesson 4](#) >> [Lesson 5](#) >> [Lesson 6](#) >> [Lesson 7](#) >> [Lesson 8](#)